

Objectives

Define the Potentiometer

How to use the Potentiometer in Arduino Projects

Change the light color of an RGB using a potentiometer

Control the LED by the potentiometer

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Notes

Comments

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Potentiometer

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Potentiometer

- A potentiometer is a three-terminal resistor with a sliding or rotating contact that forms an adjustable voltage divider.
- A potentiometer is used as a voltage divider or variable resistor in a circuit. Some applications include dimmer switches for lights, brightness controls in televisions, and faders in audio equipment.



Potentiometer

A variable resistor or Potentiometer has three main legs:

- 1- VCC
- 2- GND
- 3- Control



The VCC and GND should be connected to the Arduino terminals GND and VCC

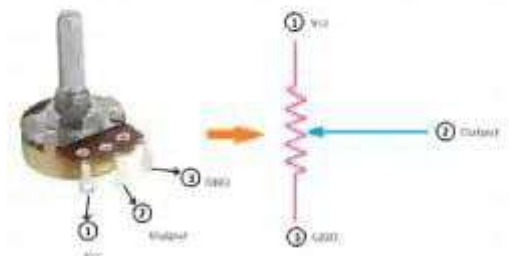
The control will return the value of the variable resistor and should be connected to one of the analog pins

Potentiometer

The potentiometer should be connected as follows:

1. The right leg should be connected to Ground port on the Arduino.
2. The left leg should be connected to 5V port on the Arduino.
3. The center leg should be connected to one of the analog ports A0 to A5.

For now, let's connect it in port A0. The potentiometer readings on the Arduino will vary between 0 and 1023.



sketch_jul22b.ino

```
1 int potentiometer=A0;
2
3 void setup() {
4   // put your setup code here, to run once:
5   Serial.begin(9600);
6 }
7
8 void loop() {
9   // put your main code here, to run repeatedly:
10  int Value=analogRead(potentiometer);
11  Serial.print("The potentiometer reading : ");
12  Serial.println(Value);
```

Output Serial Monitor X

Message (Enter to send message to Arduino UNO or COM6)

```
The potentiometer reading : 17
The potentiometer reading : 112
The potentiometer reading : 269
The potentiometer reading : 364
The potentiometer reading : 190
The potentiometer reading : 0
The potentiometer reading : 0
The potentiometer reading : 0
The potentiometer reading : 0
The potentiometer reading : 0
The potentiometer reading : 1023
The potentiometer reading : 1023
The potentiometer reading : 271
The potentiometer reading : 404
The potentiometer reading : 636
The potentiometer reading : 1023
The potentiometer reading : 1023
```

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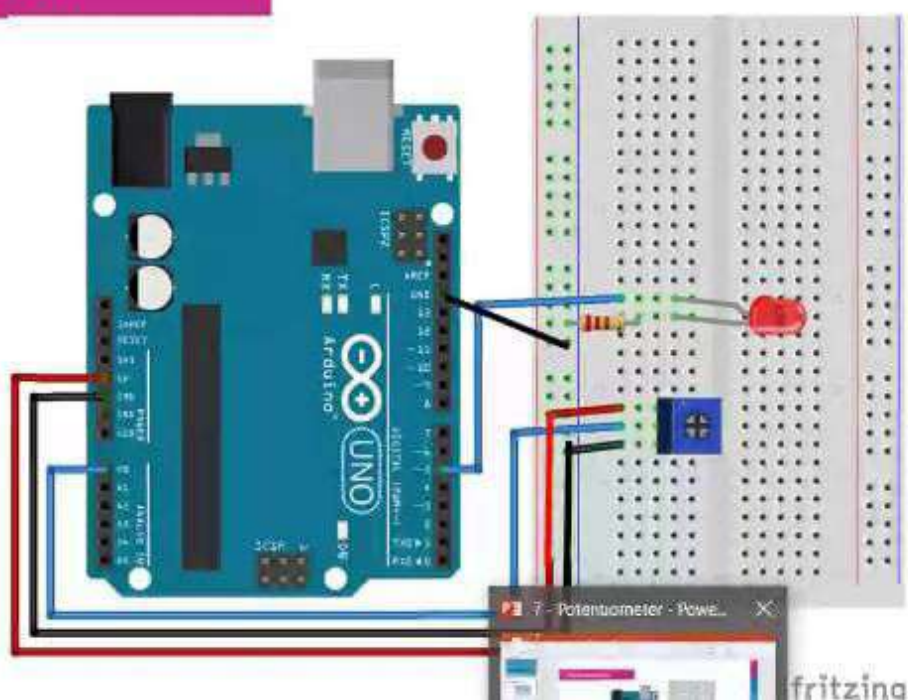
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Potentiometer

- A potentiometer is a three-terminal resistor with a sliding or rotating contact that forms an adjustable voltage divider.
- A potentiometer is used as a voltage divider or variable resistor in a circuit. Some applications include dimmer switches for lights, brightness controls in televisions, and faders in audio equipment.



Board connection



sketch_jul22c | Arduino IDE 2.3.10

File Edit Sketch Tools Help

Arduino UNO

sketch_jul22c.ino

```
3
4 void setup() {
5   // put your setup code here, to run once:
6   pinMode(led,OUTPUT);
7   Serial.begin(9600);
8 }
9
10 void loop() {
11   // put your main code here, to run repeatedly:
12   int value=analogRead(potentiometer);
13
14   if(value>=500){
15     digitalWrite(led,HIGH);
16   }else{
17     digitalWrite(led,LOW);
18   }
19   Serial.print("The potentiometer reading :");
20   Serial.println(value);
21   delay(1000);
22 }
23
```

Output - Serial Monitor X

Arduino IDE project workspace - 4/4/2021 12:07:46 PM

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More info
On 21, Jul 11 Arduino UNO on COM6

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File Edit Sketch Tools Help

Arduino UNO

sketch_jul22c.ino

```
3
4 void setup() {
5   // put your setup code here, to run once:
6   pinMode(led,OUTPUT);
7   Serial.begin(9600);
8 }
9
10 void loop() {
11   // put your main code here, to run repeatedly:
12   int value=analogRead(potentiometer);
13
14   if(value>=500){
15     digitalWrite(led,HIGH);
16   }else{
17     digitalWrite(led,LOW);
18   }
19   Serial.print("The potentiometer reading :");
20   Serial.println(value);
21   delay(1000);
22 }
23
```

Output Serial Monitor

Message (Enter to send message to Arduino UNO on COM5)

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Search the web and Windows



```
3
4 void setup() {
5   // put your setup code here, to run once:
6   pinMode(led,OUTPUT);
7   Serial.begin(9600);
8 }
9
10 void loop() {
11   // put your main code here, to run repeatedly:
12   int value=analogRead(potentiometer);
13
14   if(value>=500){
15     digitalWrite(led,HIGH);
16   }else{
17     digitalWrite(led,LOW);
18   }
19   Serial.print("The potentiometer reading :");
20   Serial.println(value);
21   delay(1000);
22 }
23
```



```
2d.ino
6  pinMode(led,OUTPUT);
7  Serial.begin(9600);
8  }
9
10 void loop() {
11     // put your main code here, to run repeatedly:
12     int value=analogRead(potentiometer);
13
14     if(value<=100){
15         digitalWrite(led,LOW);
16     }else{
17         digitalWrite(led,HIGH);
18         delay(value);
19         digitalWrite(led,LOW);
20         delay(value);
21     }
22     Serial.print("The potentiometer reading : ");
```

Output Serial Monitor X

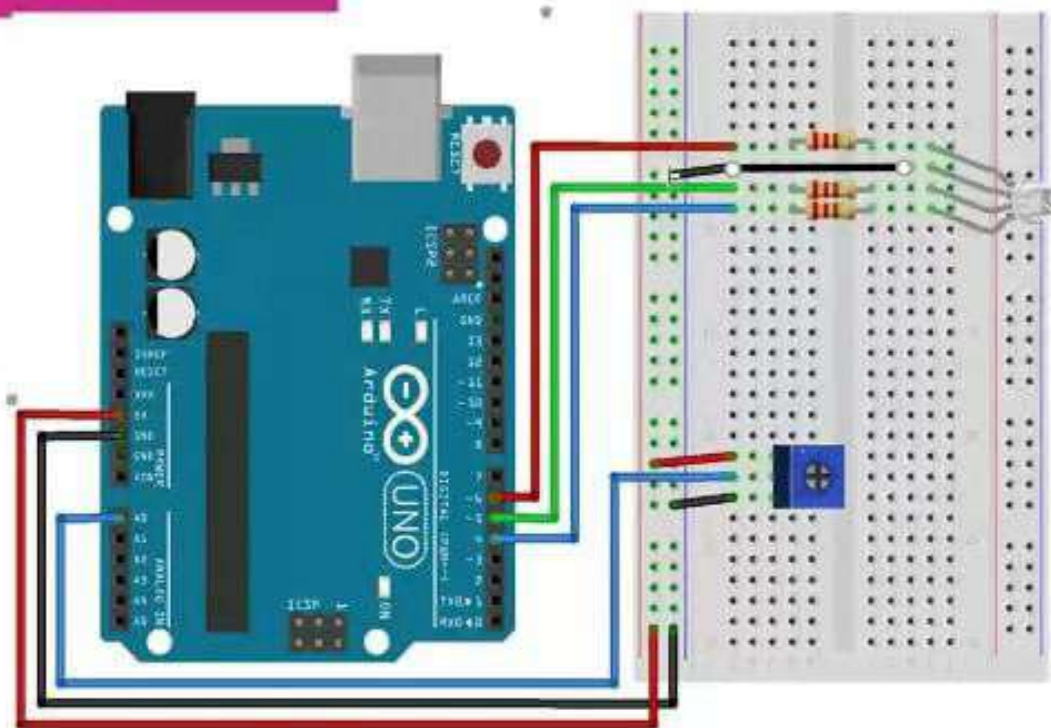
Message (Enter to send message to 'Arduino-UNO' on 'COM6')

The potentiometer reading : 788
The potentiometer reading : 797
The potentiometer reading : 483
The potentiometer reading : 478
The potentiometer reading : 0
The potentiometer reading : 0
The potentiometer reading : 0
The potentiometer reading : 393

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Board connection



```
2 int red=6;
3 int green=5;
4 int blue=4;
5
6 void setup() {
7   // put your setup code here, to run once:
8   Serial.begin(9600);
9   pinMode(red,OUTPUT);
10  pinMode(green,OUTPUT);
11  pinMode(blue,OUTPUT);
12 }
13
14 void loop() {
15   // put your main code here, to run repeatedly:
16   int value=analogRead(potentiometer);
17
18   if(value==0){
19     digitalWrite(red,LOW);
20     digitalWrite(green,LOW);
21     digitalWrite(blue,LOW);
22   }else {if(value>0 && value<300){}}
23 }
```

1

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1 of 22 - Col 53 | Arduino UNO on COM6 port connect


```

20 digitalWrite(green,LOW);
21 digitalWrite(blue,LOW);
22 }else {if(value>0 && value<300){
23 digitalWrite(red,HIGH);
24 digitalWrite(green,LOW);
25 digitalWrite(blue,LOW);
26 }else{if(value>=300 && value <600){
27 digitalWrite(red,LOW);
28 digitalWrite(green,HIGH);
29 digitalWrite(blue,LOW);
30 }else{if(value>=600 && value<900){
31 digitalWrite(red,LOW);
32 digitalWrite(green,LOW);
33 digitalWrite(blue,Blue);
34 }else{if(value>=900){
35 digitalWrite(red,HIGH);
36 digitalWrite(green,HIGH);
37 digitalWrite(blue,HIGH);digitalWrite(uint8_t pin,
38 }}}}uint8_t val) -> void
39 Serial.print("The potenti")
40 }
41

```

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sketch_jul22e.ino

```
22 }else {if(value>0 && value<300){
23     digitalWrite(red,HIGH);
24     digitalWrite(green,LOW);
25     digitalWrite(blue,LOW);
26 }else{if(value>=300 && value <600){
27     digitalWrite(red,LOW);
28     digitalWrite(green,HIGH);
29     digitalWrite(blue,LOW);
30 }else{if(value>=600 && value<900){
31     digitalWrite(red,LOW);
32     digitalWrite(green,LOW);
33     digitalWrite(blue,Blue);
34 }else{if(value>=900){
35     digitalWrite(red,HIGH);
36     digitalWrite(green,HIGH);
37     digitalWrite(blue,HIGH);
38 }}}}}
39 Serial.print("The potentiometer reading is : ");
40 Serial.println(value);
41 delay(1000);
42 }
43
```


sketch_jul22e.ino

```
21   digitalWrite(blue, LOW);
22 }else {if(value>0 && value<300){
23   digitalWrite(red, HIGH);
24   digitalWrite(green, LOW);
25   digitalWrite(blue, LOW);
26 }else{if(value>=300 && value <600){
27   digitalWrite(red, LOW);
28   digitalWrite(green, HIGH);
29   digitalWrite(blue, LOW);
30 }else{if(value>=600 && value<900){
31   digitalWrite(red, LOW);
32   digitalWrite(green, LOW);
33   digitalWrite(blue, Blue);
34 }else{if(value>=900){
35   digitalWrite(red, HIGH);
36   digitalWrite(green, HIGH);
37   digitalWrite(blue, HIGH);
```

Output

```
blue
exit status 1
```

```
Compilation error: 'Blue' was not declared in this scope
```

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stop

```
44 digitalWrite(green, LOW);
45 digitalWrite(blue, HIGH);
46 }
47
48 if(value >= 600 && value < 750){
49   digitalWrite(red, LOW);
50   digitalWrite(green, HIGH);
51   digitalWrite(blue, HIGH);
52 }
53
54 if(value >= 750 && value < 900){
55   digitalWrite(red, HIGH);
56   digitalWrite(green, HIGH);
57   digitalWrite(blue, LOW);
58 }
59
60 if(value >= 900){
61   digitalWrite(red, HIGH);
62   digitalWrite(green, HIGH);
63   digitalWrite(blue, HIGH);
64 }
65 }
```